Insect Classifier

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Registration: DF2-AI-2026-031
Task: Five-class object recognition from CIFAR-100; 32 x 32 RGB input.

Problem and dataset

The system recognises bee, beetle, butterfly, caterpillar and cockroach. A CNN is trained from scratch using only the permitted basic layers. A desktop GUI provides account registration, login, prediction, saved-image history and logout.

CIFAR-100 has 500 official training and 100 official test images per fine class. Filtering by class name gives 2,500 training-pool images and 500 test images. A stratified 80/20 split of the training pool gives 2,000 fitting images and 500 validation images. Labels are remapped in the listed order to 0 through 4.

One official test example per class



Preprocessing and augmentation

RGB conversion, resize to 32 x 32, and ToTensor scale pixels to [0,1]. Training only uses horizontal flips, rotations up to 10 degrees and translations up to 8%. Validation, test and GUI inference use deterministic preprocessing.

Source: Krizhevsky (2009), Learning Multiple Layers of Features from Tiny Images.
www.cs.toronto.edu/~kriz/cifar.html
Dataset files are downloaded automatically and excluded from the submitted ZIP.

CNN architecture

Layer / operation	Parameters	Output (excluding batch)
Input	RGB, normalised [0,1]	3 x 32 x 32
Conv2d + ReLU	3 -> 32, kernel 3, padding 1	32 x 32 x 32
MaxPool2d	kernel 2, stride 2	32 x 16 x 16
Conv2d + ReLU	32 -> 64, kernel 3, padding 1	64 x 16 x 16
MaxPool2d	kernel 2, stride 2	64 x 8 x 8
Conv2d + ReLU	64 -> 128, kernel 3, padding 1	128 x 8 x 8
MaxPool2d	kernel 2, stride 2	128 x 4 x 4
Flatten	128 * 4 * 4	2048
Linear + ReLU	2048 -> 128	128
Dropout	p = 0.4 (training only)	128
Output Linear	128 -> 5 raw logits	5

Why these layers?

Convolution filters learn local edge, texture and shape patterns. ReLU introduces nonlinearity. Pooling reduces spatial size. The hidden Linear layer combines learned features, and Dropout regularises the classifier. Three convolution blocks exceed the required minimum of two.

No pretrained model, transfer learning, BatchNorm or adaptive pooling is used. CrossEntropyLoss receives raw logits; softmax is applied only for inference confidence. The network has 356,165 trainable parameters.

Training and learning curves

Adam optimiser; learning rate 0.001; weight decay 0.0001; batch size 64; CrossEntropyLoss; random seed 42. The checkpoint is selected by validation accuracy. Train/validation/test metrics are printed each epoch, but test results are not used to choose the checkpoint or tune settings.

Measured run: 30 epochs on cpu; selected epoch 29; elapsed 43.8 seconds. Split sizes: 2000 train / 500 validation / 500 test.



Interpreting the curves

Training loss measures fit to augmented training images. Validation loss and accuracy measure generalisation on held-out training-pool images. A widening train/validation gap or rising validation loss while training improves is evidence consistent with overfitting. Test curves are monitoring evidence only.

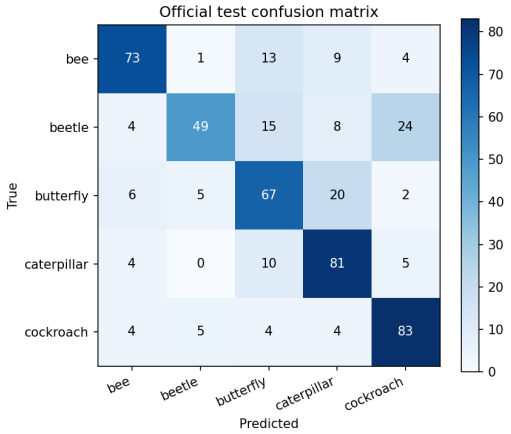
The saved state_dict contains learned weights; class order, input size and normalisation metadata accompany it. The GUI loads the checkpoint in evaluation mode, so Dropout is disabled. It never retrains during startup.

Evaluation and error analysis

Official held-out test accuracy: 70.60% on 500 images. Per-class values are recalls (correct examples divided by all true examples in that class).

Class	Per-class accuracy
bee	73.00%
beetle	49.00%
butterfly	67.00%
caterpillar	81.00%
cockroach	83.00%

Cockroach performs best (83%); beetle is weakest (49%). Of 100 true beetle images, 24 are predicted as cockroach. Shared visual cues are one possible explanation, not an established cause.





bee mistaken for beetle: Possible small-image ambiguity, shared colour/texture or background cues. This visual interpretation is a hypothesis, not a proven causal explanation.

bee mistaken for cockroach: Possible small-image ambiguity, shared colour/texture or background cues. This visual interpretation is a hypothesis, not a proven causal explanation.

bee mistaken for cockroach: Possible small-image ambiguity, shared colour/texture or background cues. This visual interpretation is a hypothesis, not a proven causal explanation.

Database design and security

Relational schema: one user to many saved images

users	image_records
id INTEGER PRIMARY KEY	id INTEGER PRIMARY KEY
username TEXT UNIQUE NOT NULL	user_id INTEGER FK -> users.id
salt TEXT NOT NULL	image_path TEXT NOT NULL
password_hash TEXT NOT NULL	predicted_label TEXT NOT NULL
created_at TEXT NOT NULL	confidence REAL NOT NULL
	true_label TEXT NULL; created_at TEXT

A new SQLite database is created automatically. Passwords are salted with 16 cryptographically random bytes and hashed using PBKDF2-HMAC-SHA256 with 600,000 iterations. Login recomputes the hash and uses constant-time comparison. Passwords are never stored as plain text.

All SQL input uses bound parameters. History and deletion queries include the logged-in user ID. Foreign keys are enabled. The app copies each saved image as a uniquely named RGB PNG into app_data/records; deleting a record removes the app copy, not the original source file.

Workflow and tested behaviour

Register / login -> open image -> preview and top-3 prediction -> optional true label -> Save -> user history -> preview/delete/CSV export -> logout. Missing models, invalid images and wrong credentials produce clear dialogs.

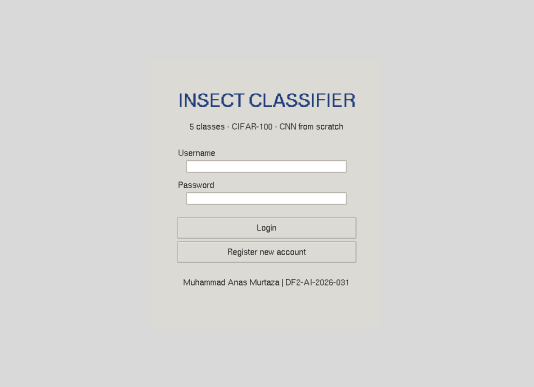
Automated checks cover salted authentication, duplicate usernames, SQL-like password input, cross-user record isolation, ownership-protected deletion, copied-image cleanup, permitted layer types, tensor shape, backward propagation, checkpoint loading and probability normalisation.

Limits

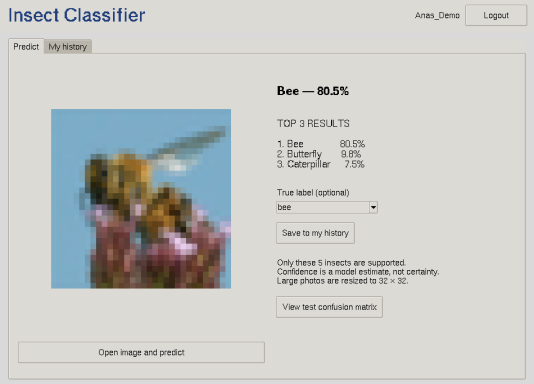
The model always chooses one of the five classes, including for out-of-scope photos. Confidence is uncalibrated. Resizing to 32 x 32 loses detail, and personal photos may differ from CIFAR. SQLite is local and not encrypted; filesystem permissions still matter.

Desktop application and conclusion

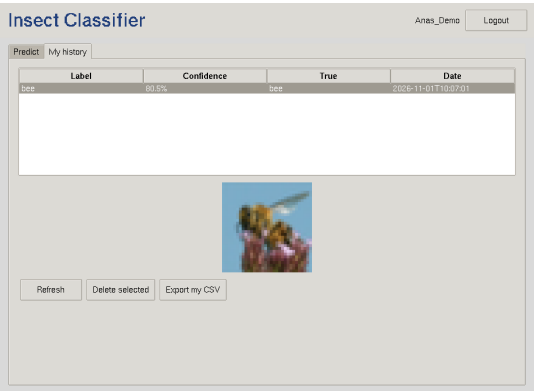
1. Login / registration



2. Prediction, confidence and top-3



3. Saved history with preview



The application connects a basic CNN to secure user accounts and per-user image storage. Future work could improve validation-based tuning, calibration and performance on realistic insect photos. Review measured errors and be ready to explain every module during the individual viva.